Chapter 16
JavaFX UI Controls and Multimedia



Liang, Introduction to Java Programming, Eleventh Edition, (c) 2018 Pearson Education, Ltd. All rights reserved.

Motivations

- A graphical user interface (GUI) makes a system user-friendly and easy to use. Creating a GUI requires creativity and knowledge of how GUI components work. Since the GUI components in Java are very flexible and versatile, you can create a wide assortment of useful user interfaces.
- Previous chapters briefly introduced several GUI components. This chapter introduces the frequently used GUI components in detail.



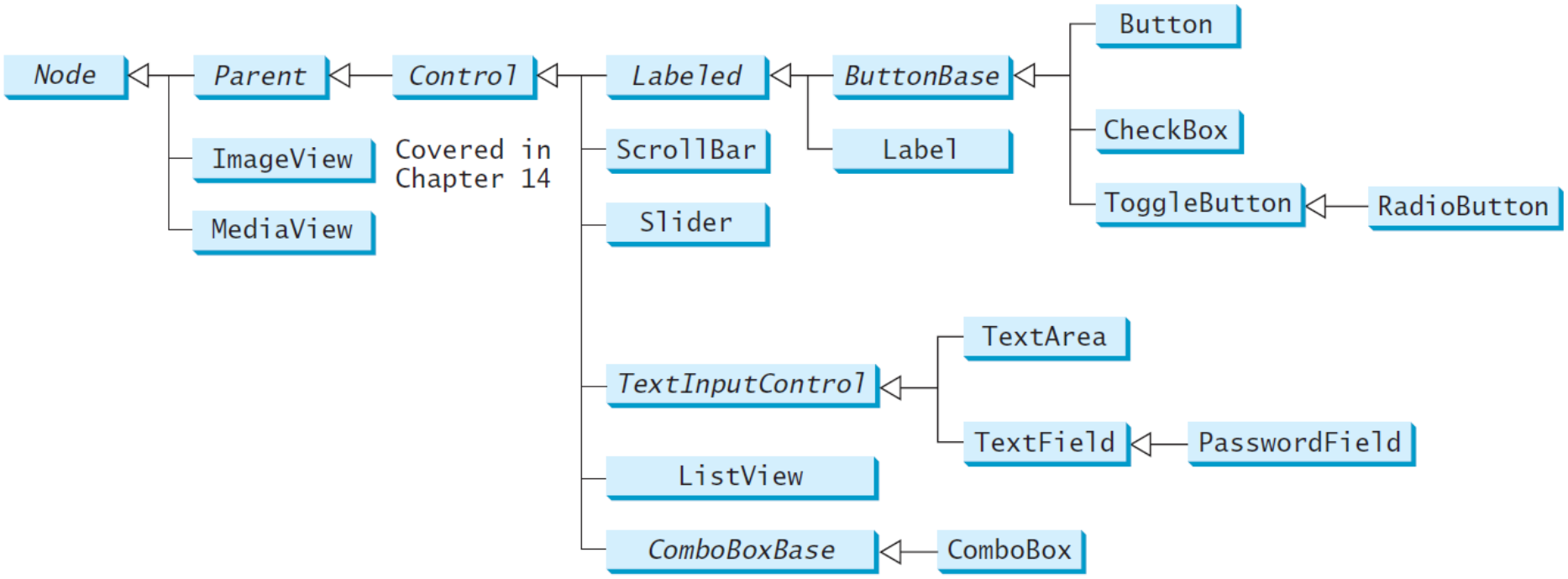
Objectives

- To create graphical user interfaces with various user-interface controls (§§16.2–16.11).
- To create a label with text and graphic using the **Label** class and explore properties in the abstract **Labeled** class (§16.2).
- To create a button with text and graphic using the **Button** class and set a handler using the **setOnAction** method in the abstract **ButtonBase** class (§16.3).
- To create a check box using the **CheckBox** class (§16.4).
- To create a radio button using the **RadioButton** class and group radio buttons using a **ToggleGroup** (§16.5).
- To enter data using the **TextField** class and password using the **PasswordField** class (§16.6).
- To enter data in multiple lines using the **TextArea** class (§16.7).
- To select a single item using **ComboBox** (§16.8).
- To select a single or multiple items using **ListView** (§16.9).
- To select a range of values using **ScrollBar** (§16.10).
- To select a range of values using **Slider** and explore differences between **ScrollBar** and **Slider** (§16.11).
- To develop a tic-tac-toe game (§16.12).
- To view and play video and audio using the **Media**, **MediaPlayer**, and **MediaView** (§16.13).
- To develop a case study for showing the national flag and play anthem (§16.14).



Liang, Introduction to Java Programming, Eleventh Edition, (c) 2018 Pearson Education, Ltd. All rights reserved.

Frequently Used UI Controls

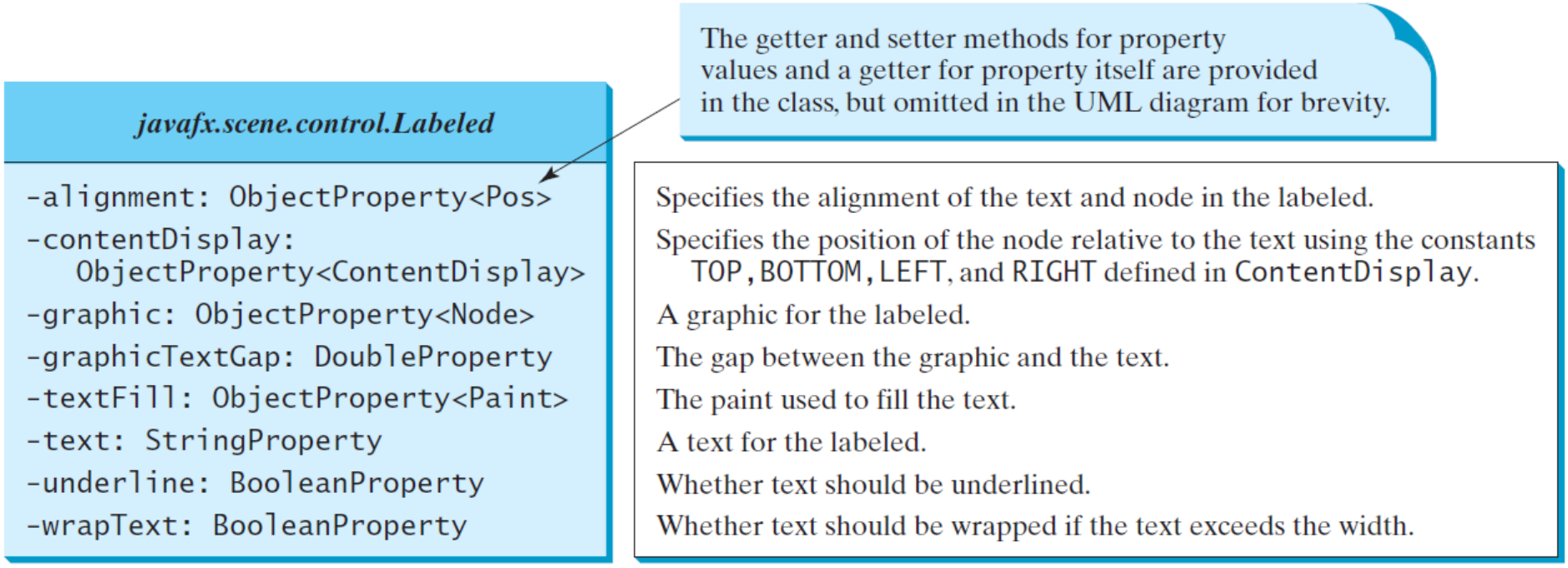


Throughout this book, the prefixes **lbl**, **bt**, **chk**, **rb**, **tf**, **pf**, **ta**, **cbo**, **lv**, **scb**, **sld**, and **mp** are used to name reference variables for **Label**, **Button**, **CheckBox**, **RadioButton**, **TextField**, **PasswordField**, **TextArea**, **ComboBox**, **ListView**, **ScrollBar**, **Slider**, and **MediaPlayer**.

Liang, Introduction to Java Programming, Eleventh Edition, (c) 2018 Pearson Education, Ltd. All rights reserved.

Labeled

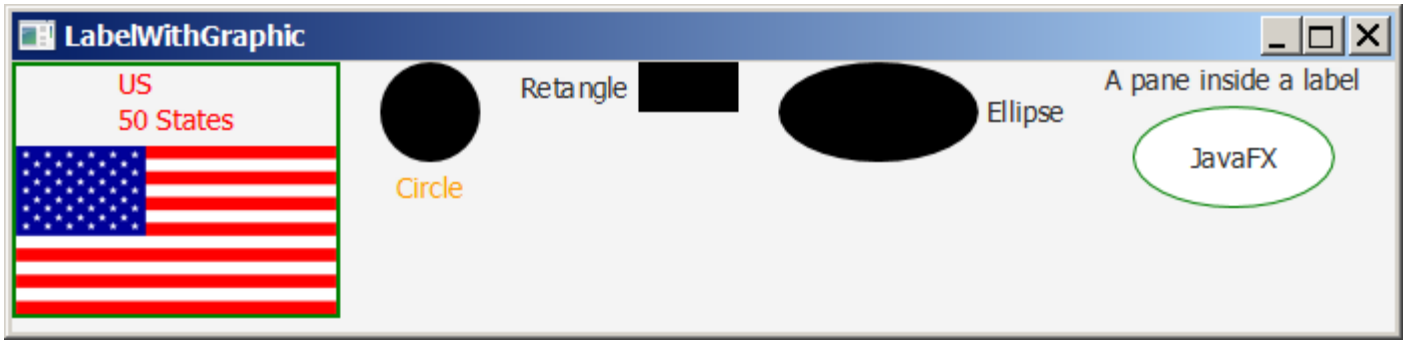
A *label* is a display area for a short text, a node, or both. It is often used to label other controls (usually text fields). Labels and buttons share many common properties. These common properties are defined in the **Labeled** class.

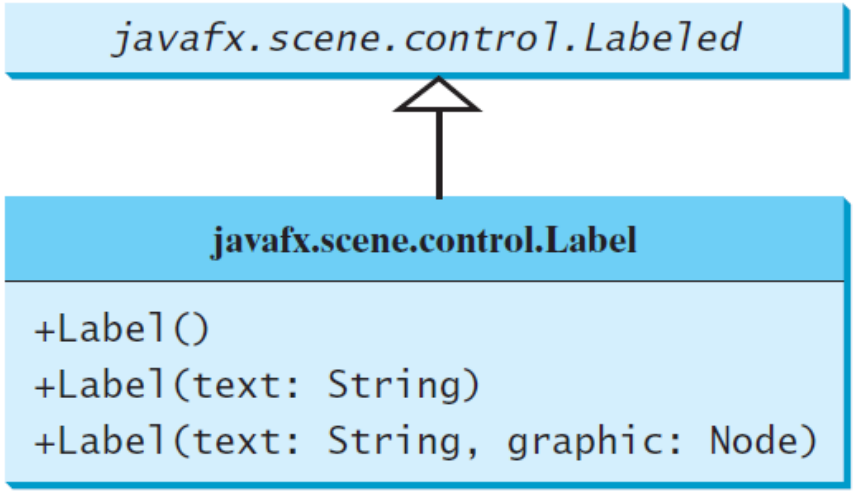


Liang, Introduction to Java Programming, Eleventh Edition, (c) 2018 Pearson Education, Ltd. All rights reserved.

Label

The Label class defines labels.





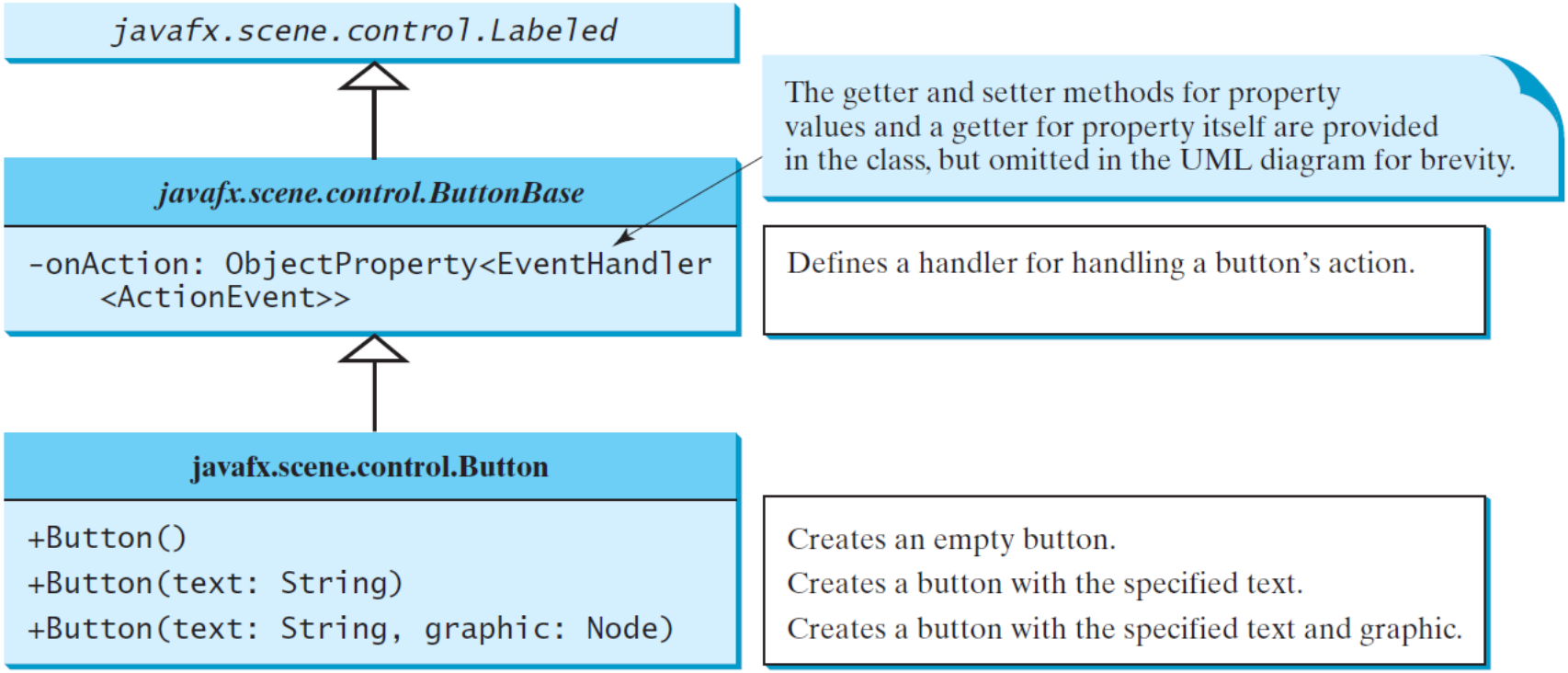
Creates an empty label.
Creates a label with the specified text.
Creates a label with the specified text and graphic.

LabelWithGraphic Run

Liang, Introduction to Java Programming, Eleventh Edition, (c) 2018 Pearson Education, Ltd. All rights reserved.

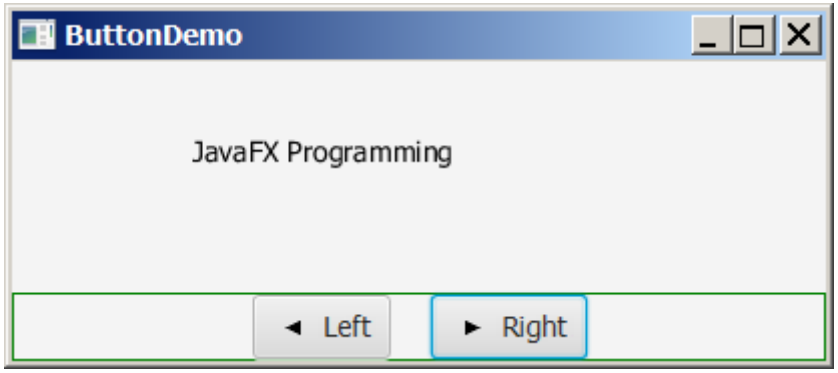
ButtonBase and Button

A *button* is a control that triggers an action event when clicked. JavaFX provides regular buttons, toggle buttons, check box buttons, and radio buttons. The common features of these buttons are defined in **ButtonBase** and **Labeled** classes.



Liang, Introduction to Java Programming, Eleventh Edition, (c) 2018 Pearson Education, Ltd. All rights reserved.

Button Example

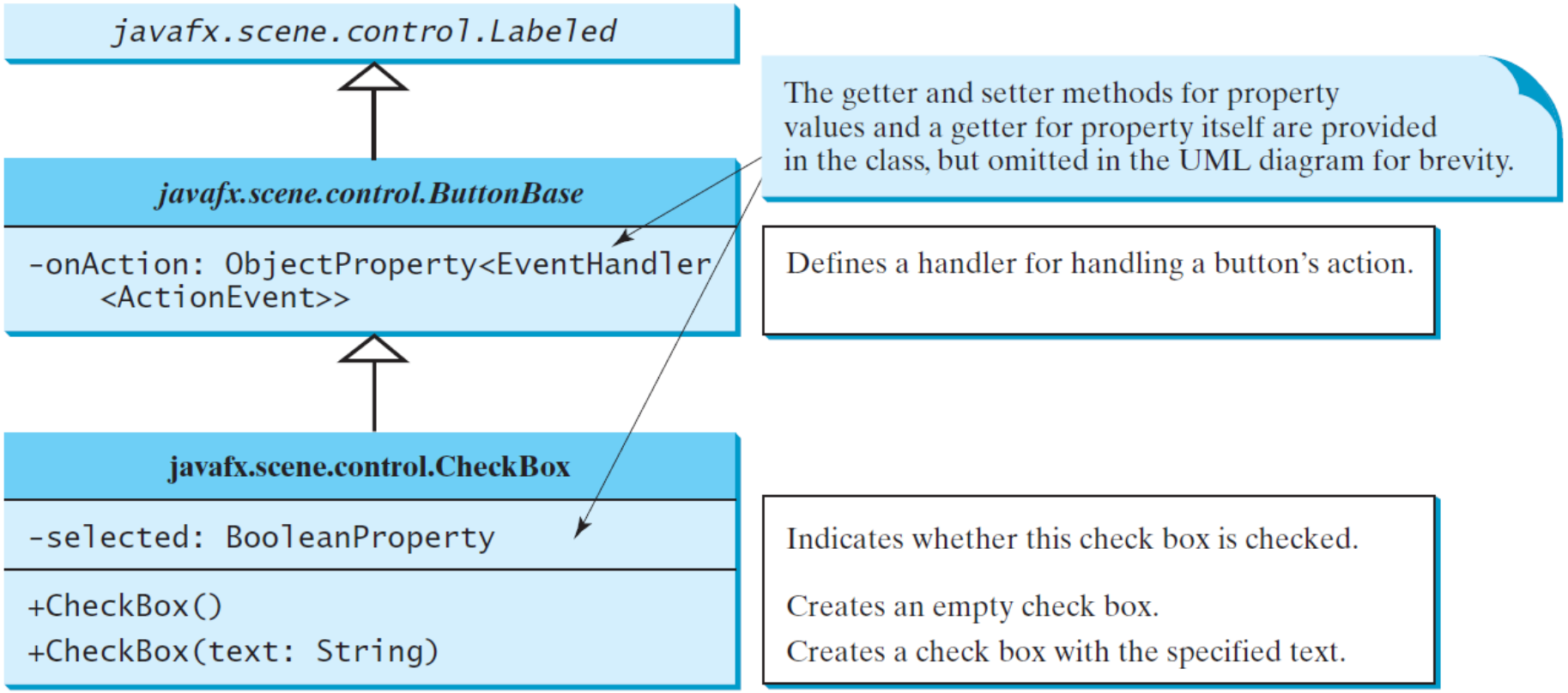


ButtonDemo Run

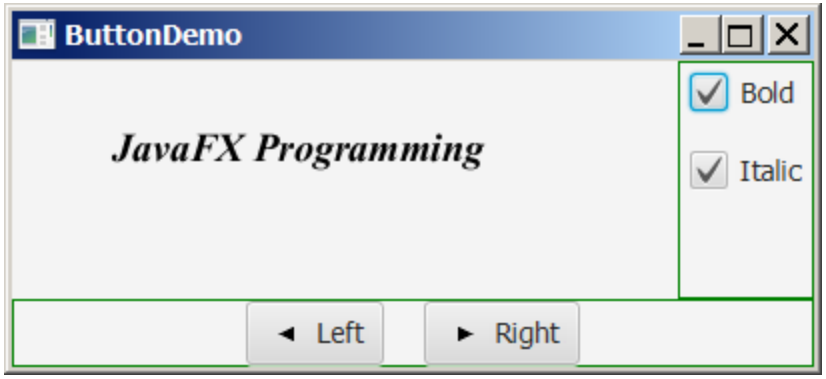
Liang, Introduction to Java Programming, Eleventh Edition, (c) 2018 Pearson Education, Ltd. All rights reserved.

CheckBox

A **CheckBox** is used for the user to make a selection. Like **Button**, **CheckBox** inherits all the properties such as **onAction**, **text**, **graphic**, **alignment**, **graphicTextGap**, **textFill**, **contentDisplay** from **ButtonBase** and **Labeled**.



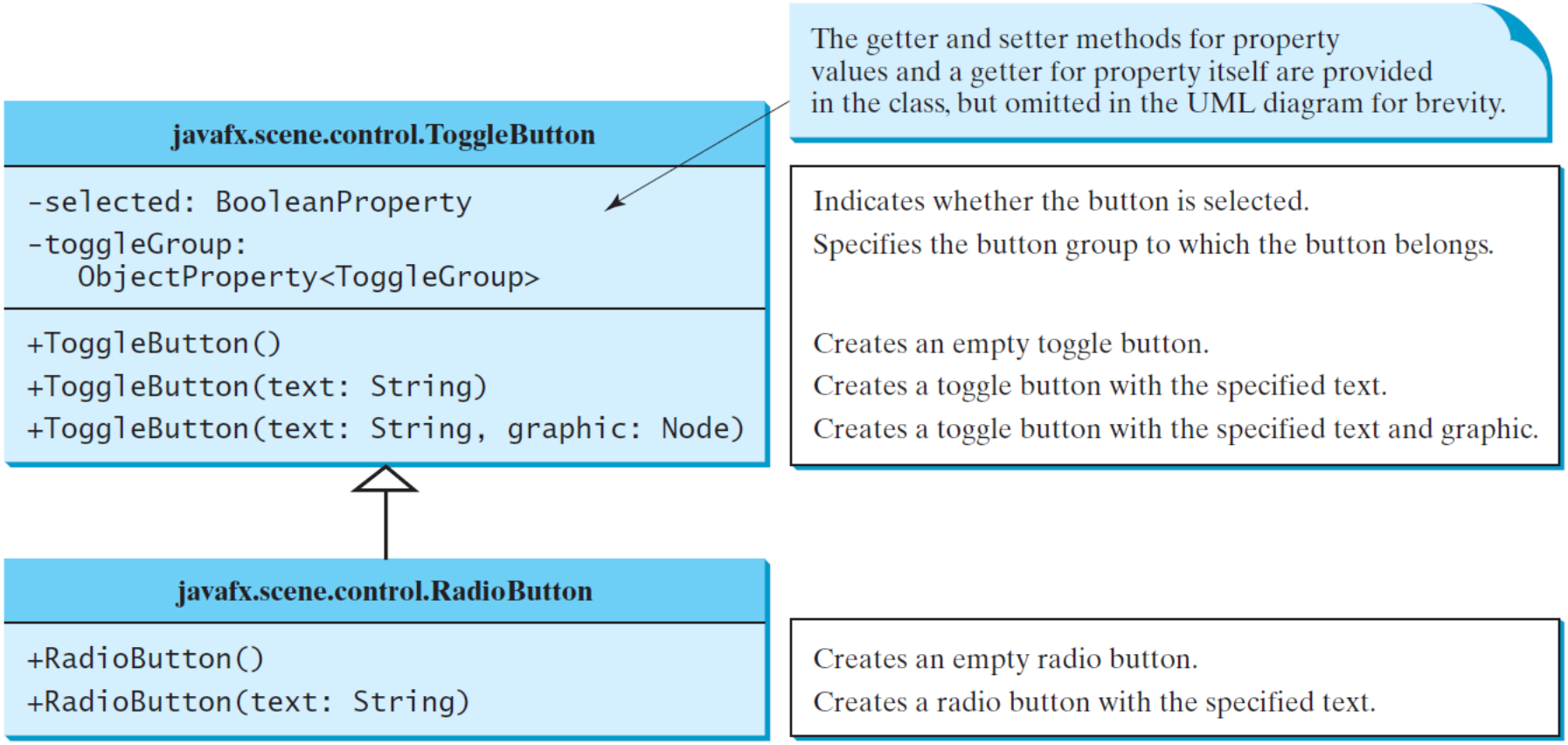
CheckBox Example



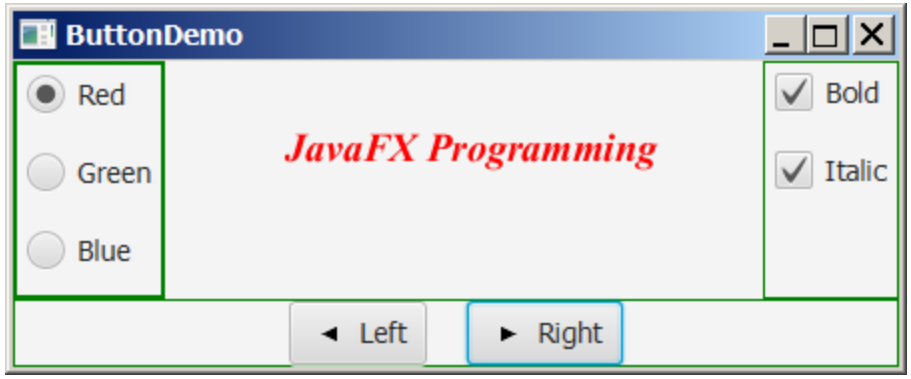
CheckBoxDemo Run

RadioButton

Radio buttons, also known as *option buttons*, enable you to choose a single item from a group of choices. In appearance radio buttons resemble check boxes, but check boxes display a square that is



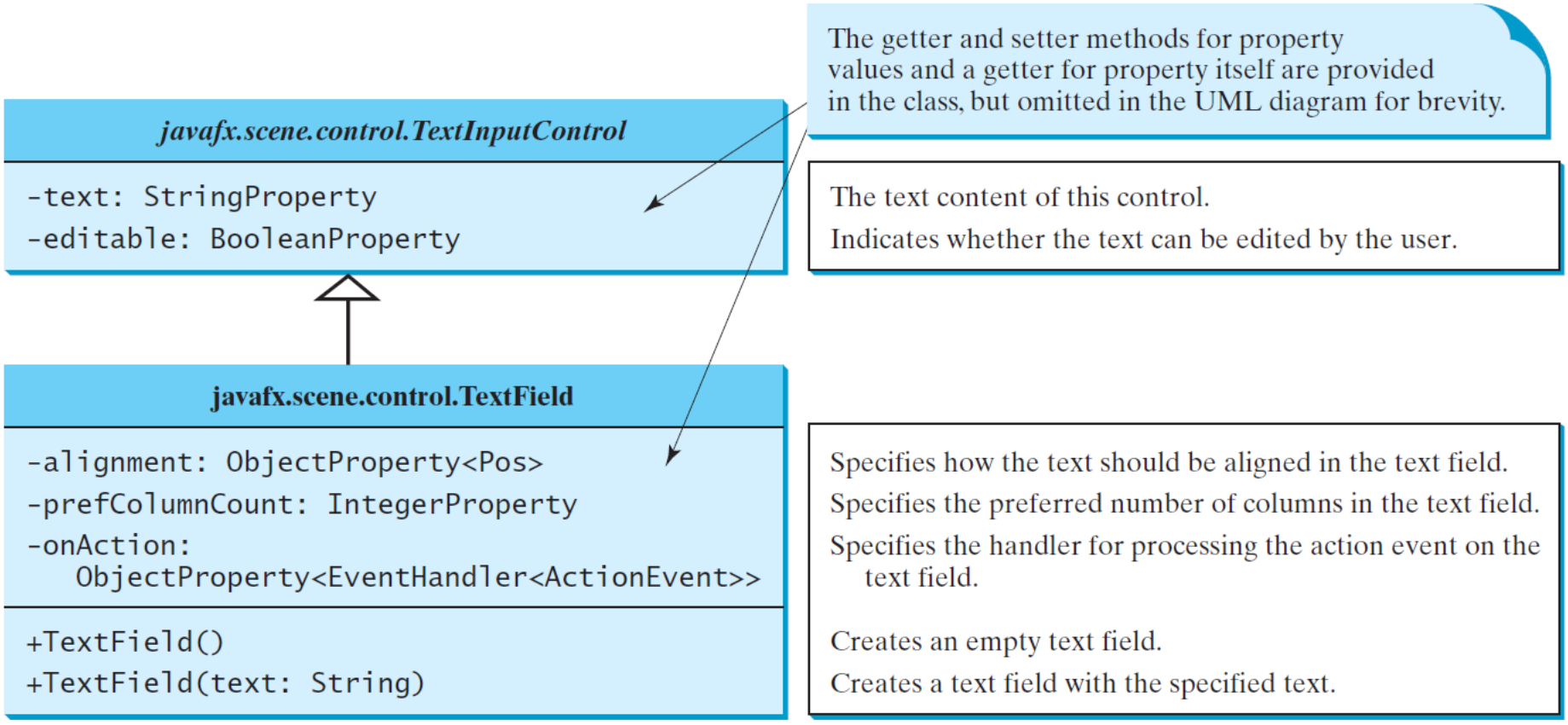
RadioButton Example



RadioButtonDemo Run

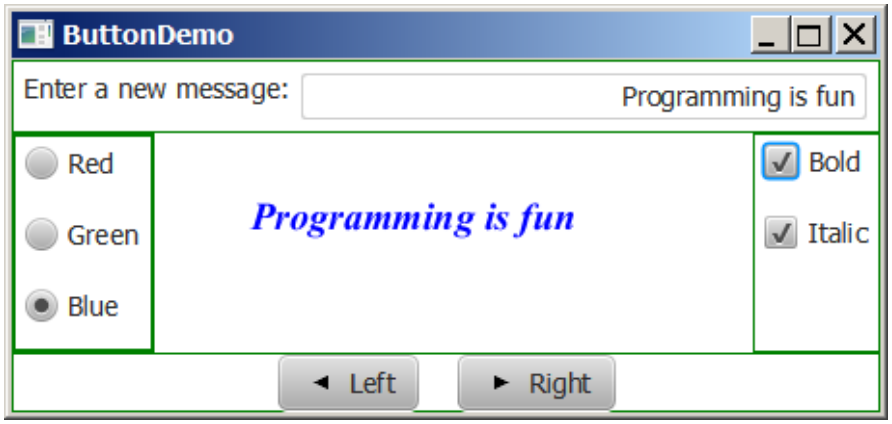
TextField

A text field can be used to enter or display a string. **TextField** is a subclass of **TextInputControl**.



Liang, Introduction to Java Programming, Eleventh Edition, (c) 2018 Pearson Education, Ltd. All rights reserved.

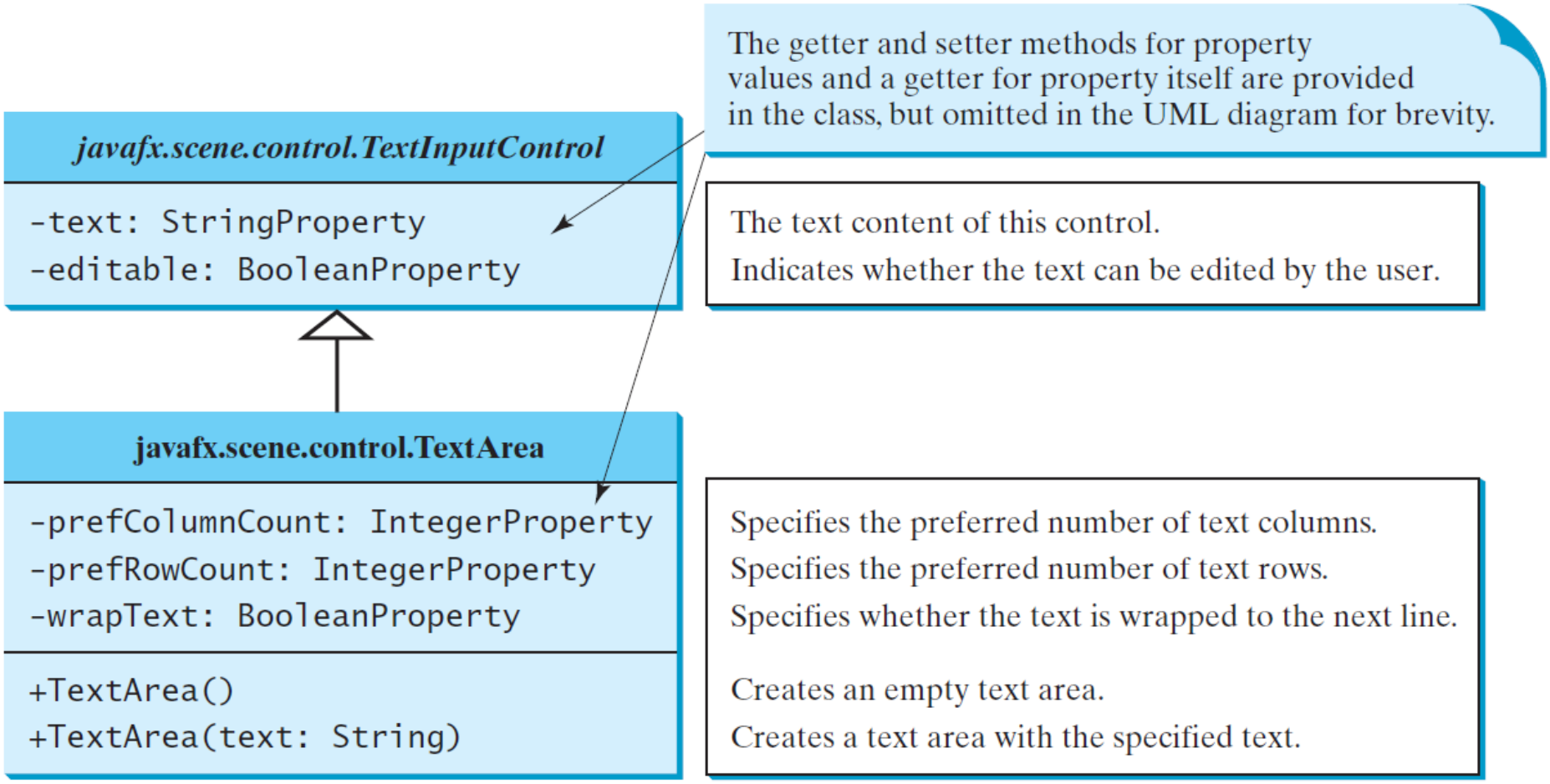
TextField Example



TextFieldDemo Run

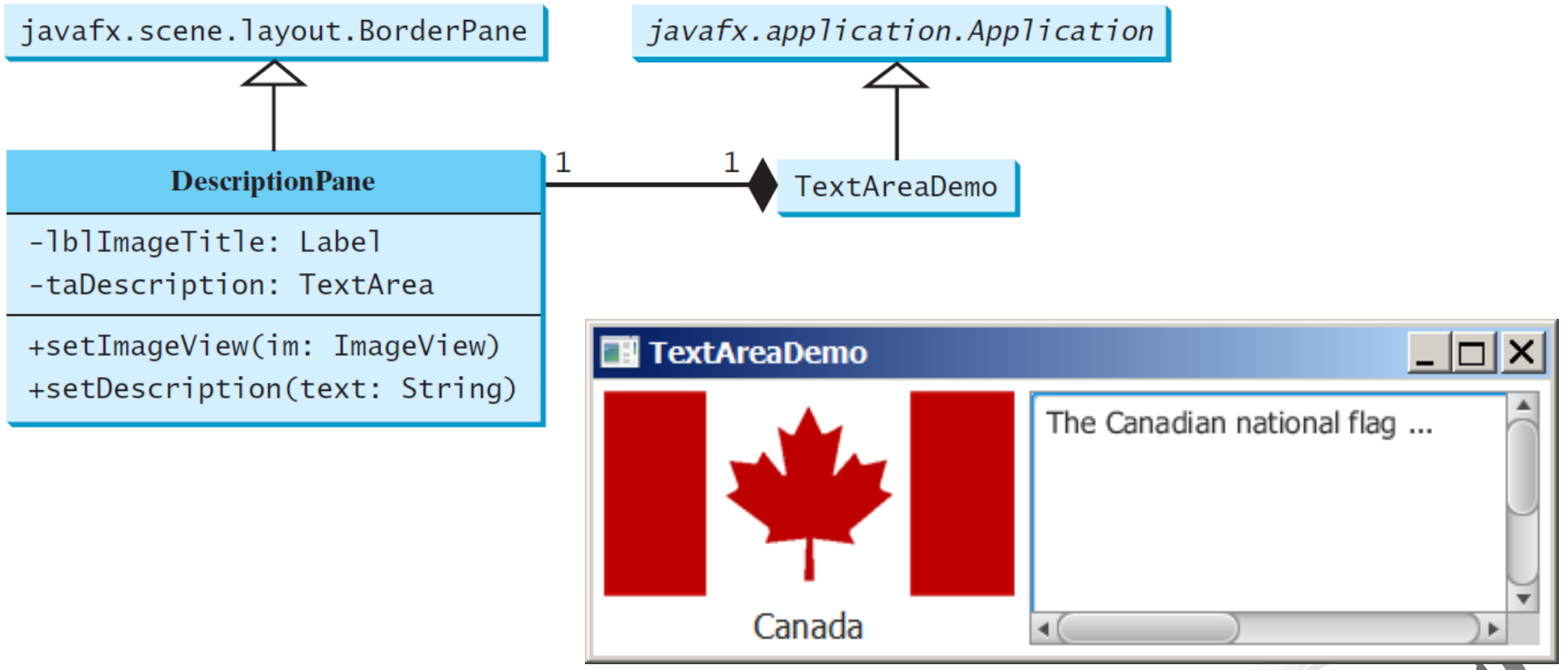
TextArea

A **TextArea** enables the user to enter multiple lines of text.



Liang, Introduction to Java Programming, Eleventh Edition, (c) 2018 Pearson Education, Ltd. All rights reserved.

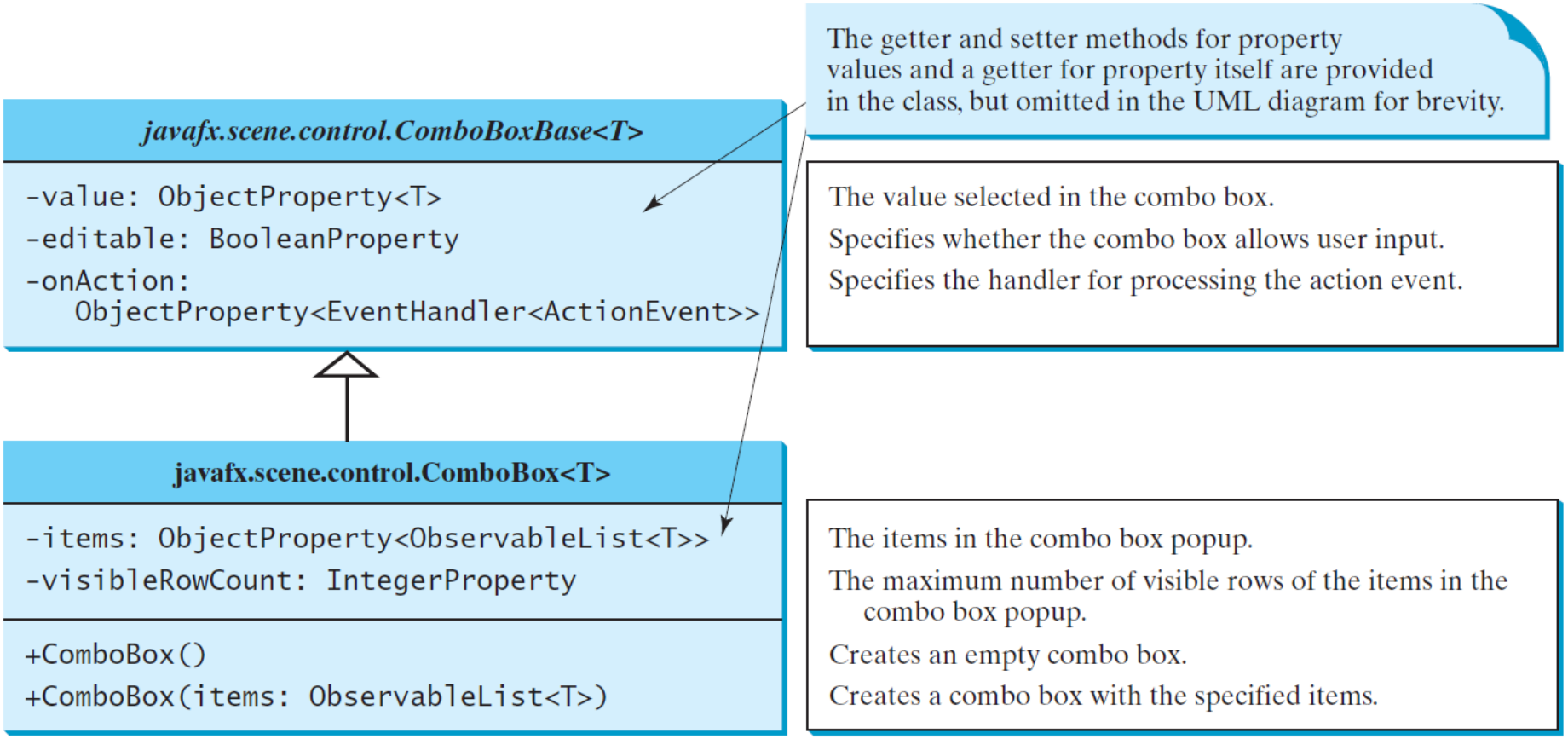
TextArea Example



DescriptionPane TextAreaDemo Run

ComboBox

A combo box, also known as a choice list or drop-down list, contains a list of items from which the user can choose.



ComboBox Example

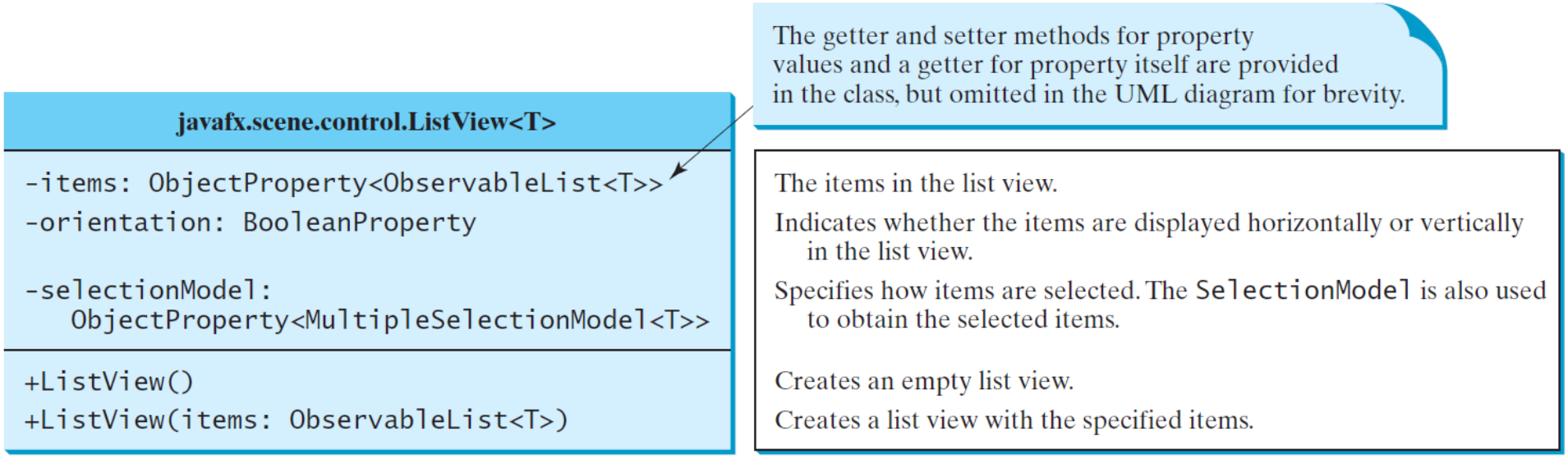
This example lets users view an image and a description of a country's flag by selecting the country from a combo box.



ComboBoxDemo Run

ListView

A *list view* is a component that performs basically the same function as a combo box, but it enables the user to choose a single value or multiple values.



Example: Using ListView

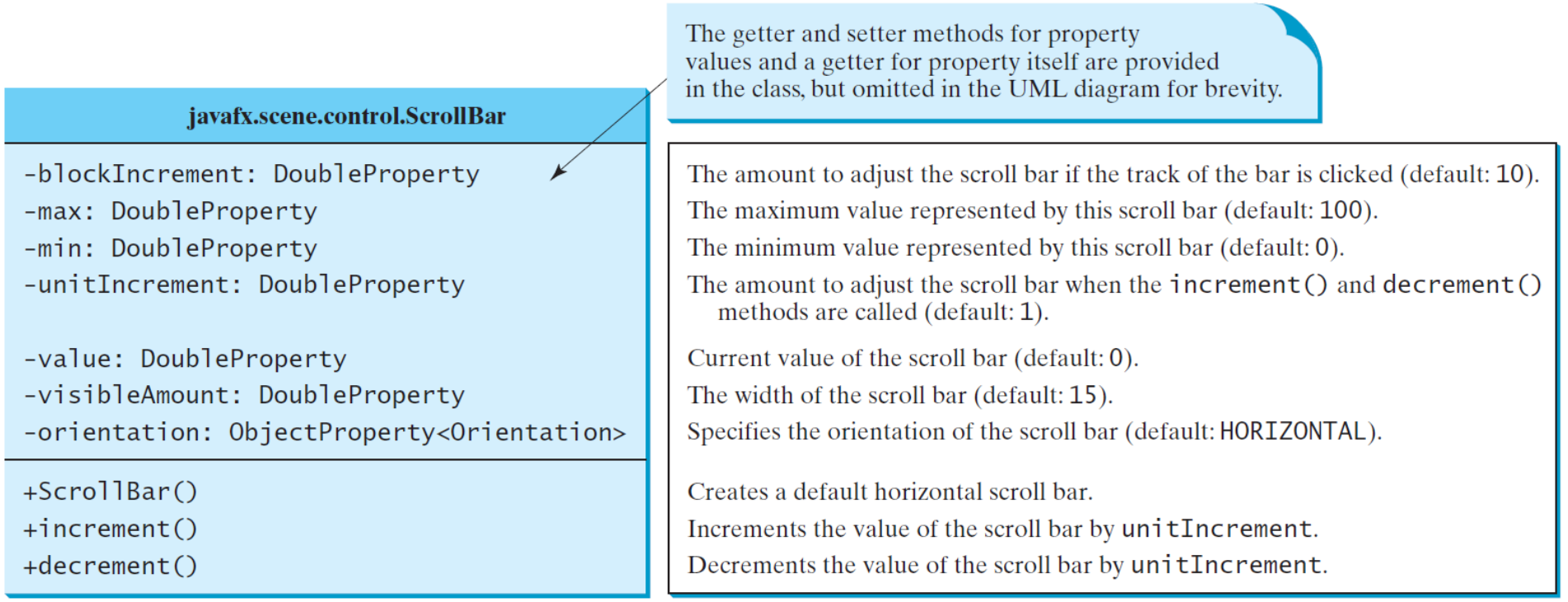
This example gives a program that lets users select countries in a list and display the flags of the selected countries in the labels.



ListViewDemo Run

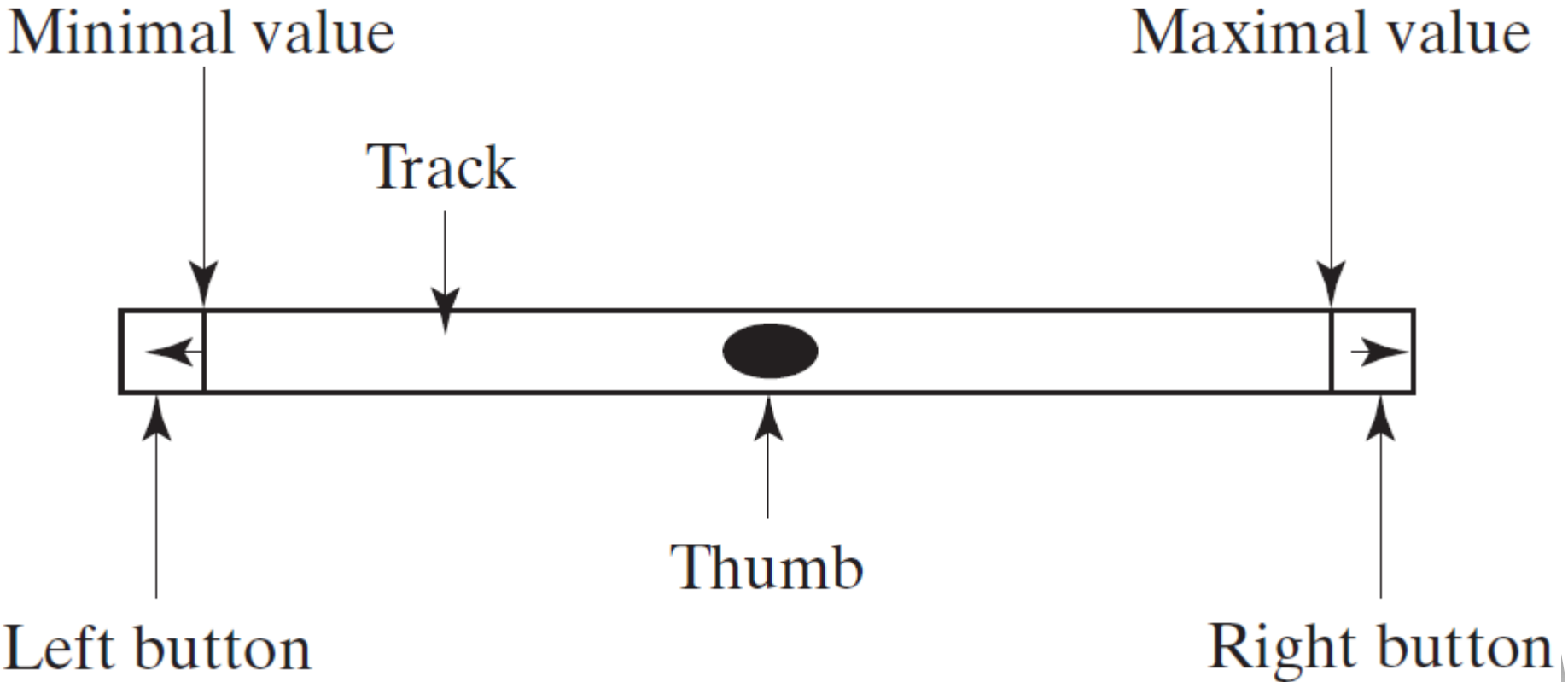
ScrollBar

A *scroll bar* is a control that enables the user to select from a range of values. The scrollbar appears in two styles: *horizontal* and *vertical*.



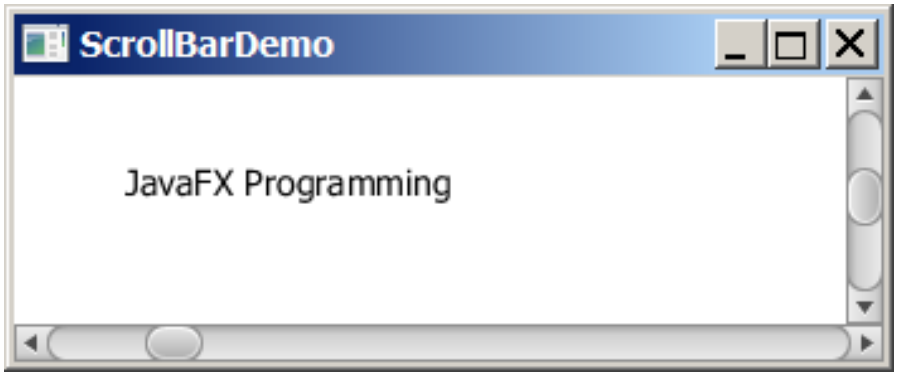
Liang, Introduction to Java Programming, Eleventh Edition, (c) 2018 Pearson Education, Ltd. All rights reserved.

Scroll Bar Properties



Example: Using Scrollbars

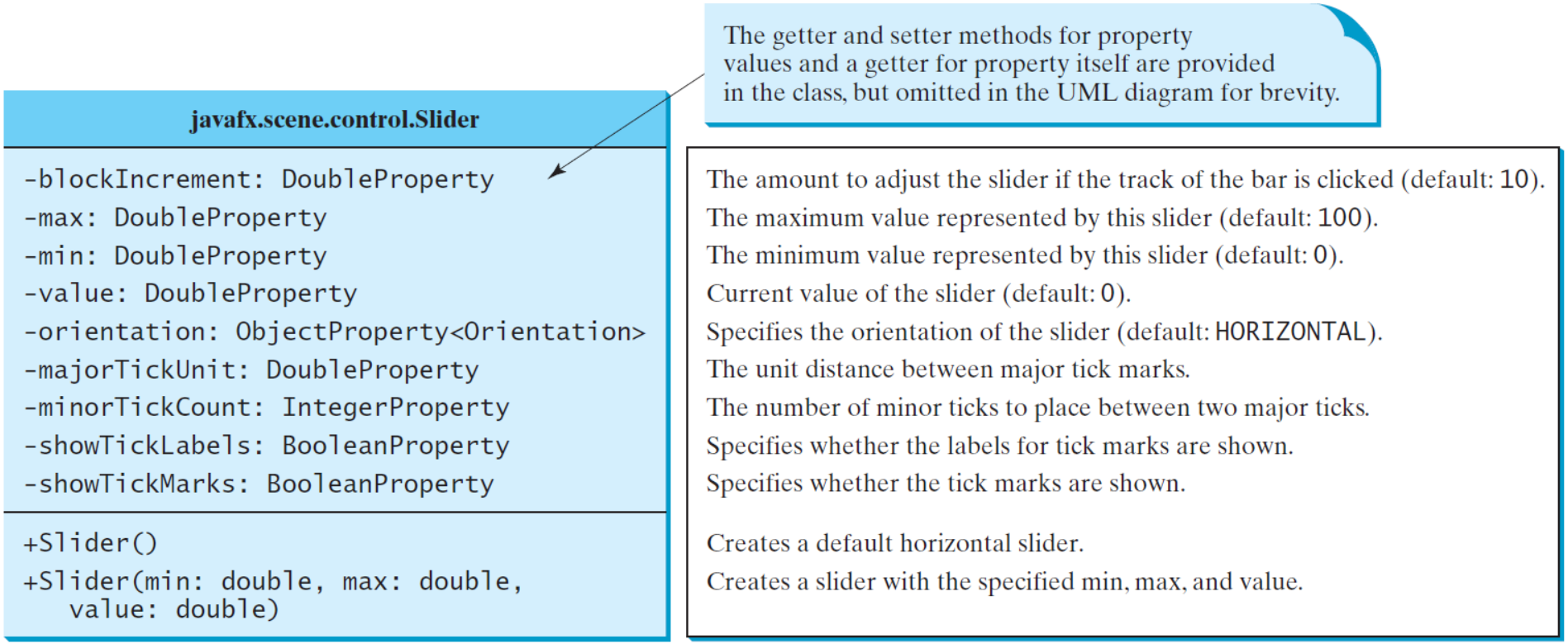
This example uses horizontal and vertical scrollbars to control a message displayed on a panel. The horizontal scrollbar is used to move the message to the left or the right, and the vertical scrollbar to move it up and down.



ScrollBarDemo Run

Slider

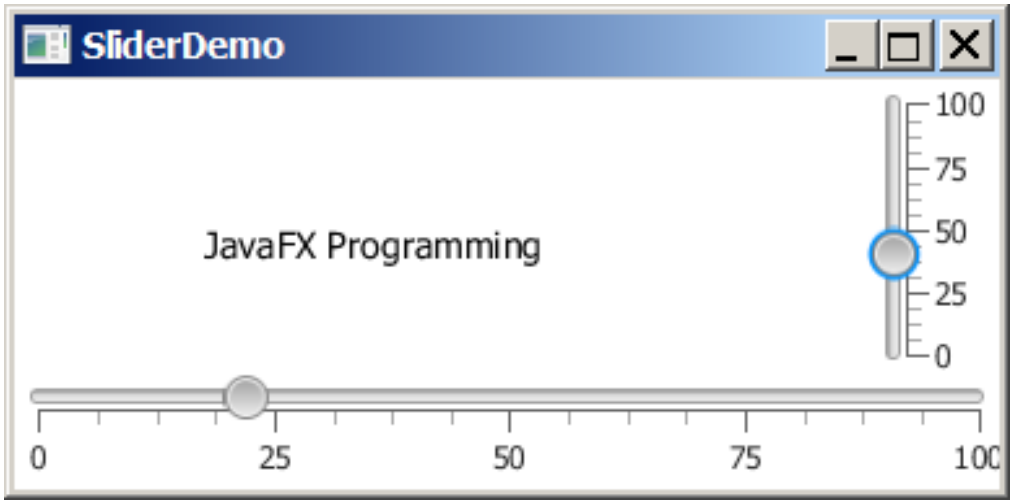
Slider is similar to ScrollBar, but Slider has more properties and can appear in many forms.



Liang, Introduction to Java Programming, Eleventh Edition, (c) 2018 Pearson Education, Ltd. All rights reserved.

Example: Using Sliders

Rewrite the preceding program using the sliders to control a message displayed on a panel instead of using scroll bars.

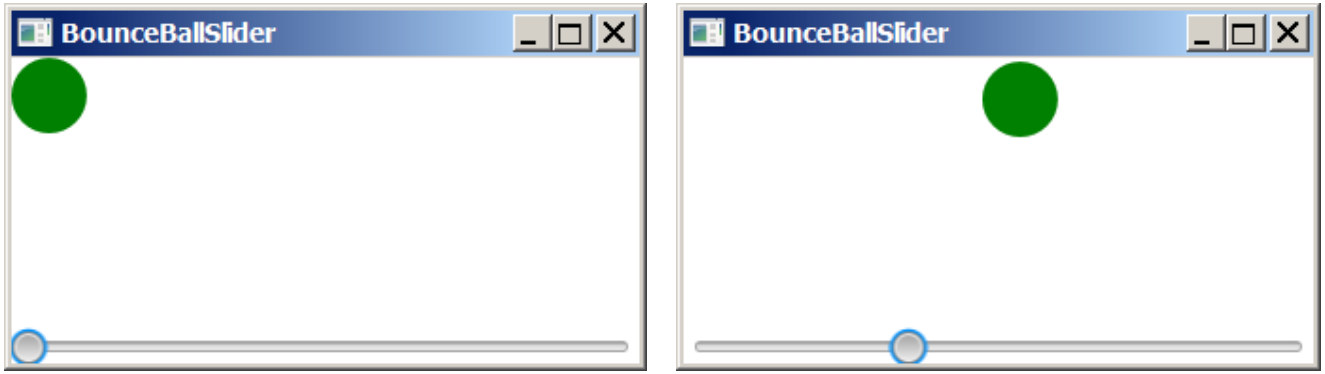




Liang, Introduction to Java Programming, Eleventh Edition, (c) 2018 Pearson Education, Ltd. All rights reserved.

Case Study: Bounce Ball

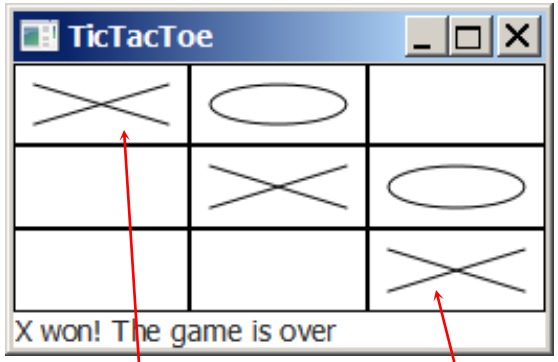
Listing 15.17 gives a program that displays a bouncing ball. You can add a slider to control the speed of the ball movement.

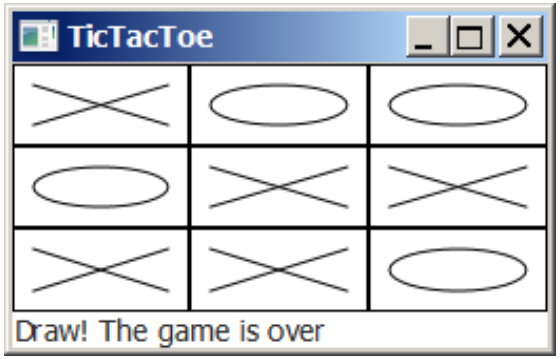


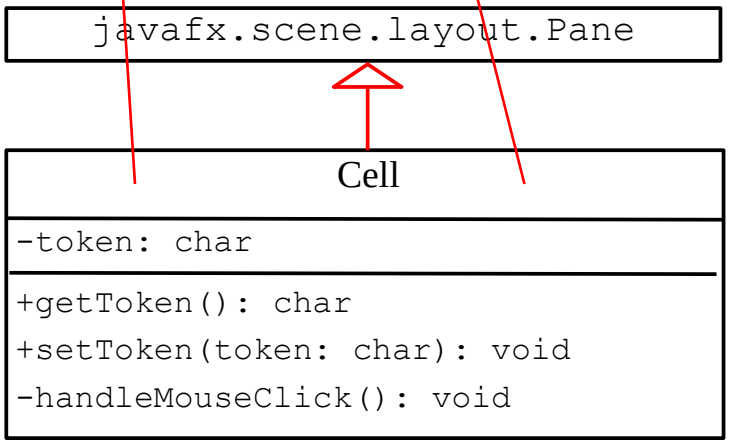
SliderDemo Run

Liang, Introduction to Java Programming, Eleventh Edition, (c) 2018 Pearson Education, Ltd. All rights reserved.

Case Study: TicTacToe







Token used in the cell (default: ' ').

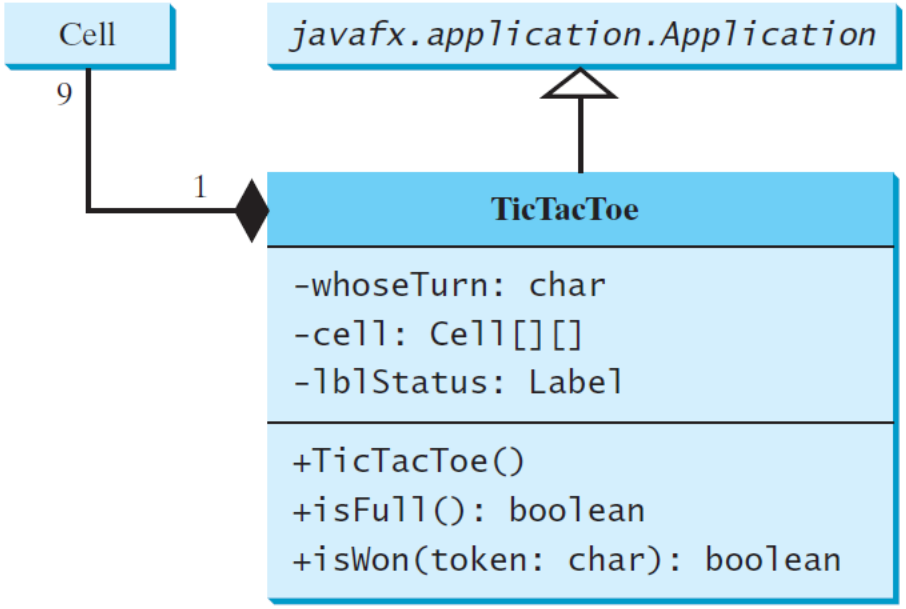
Returns the token in the cell.

Sets a new token in the cell.

Handles a mouse click event.

Liang, Introduction to Java Programming, Eleventh Edition, (c) 2018 Pearson Education, Ltd. All rights reserved.

Case Study: TicTacToe, cont.



Indicates which player has the turn, initially X.
A 3 × 3, two-dimensional array for cells.
A label to display game status.

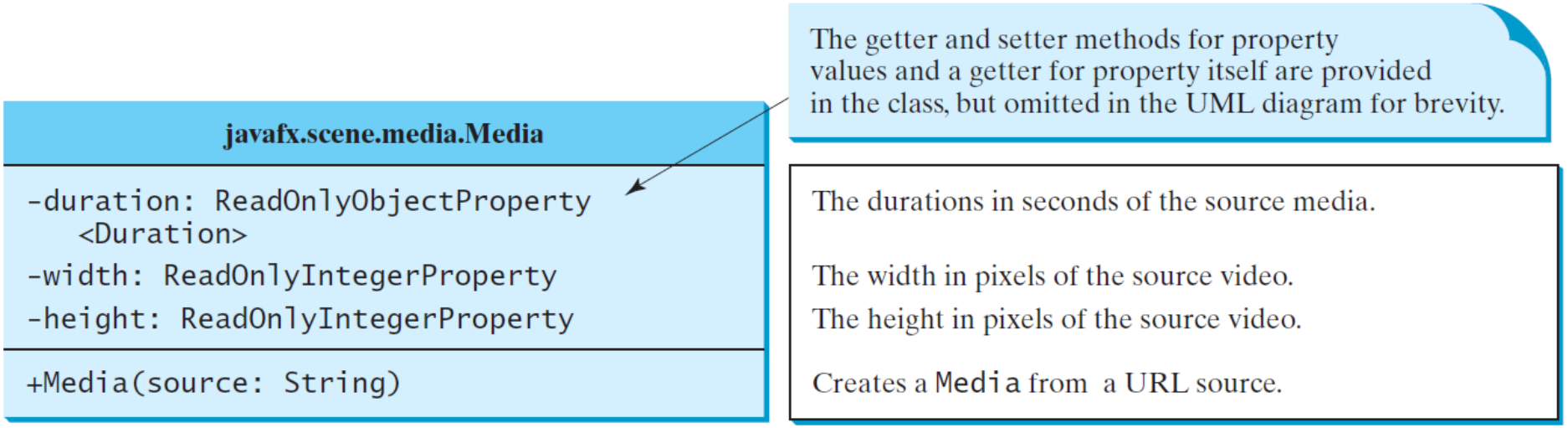
Constructs the TicTacToe user interface.
Returns true if all cells are filled.
Returns true if a player with the specified token has won.

TicTacToe Run

Liang, Introduction to Java Programming, Eleventh Edition, (c) 2018 Pearson Education, Ltd. All rights reserved.

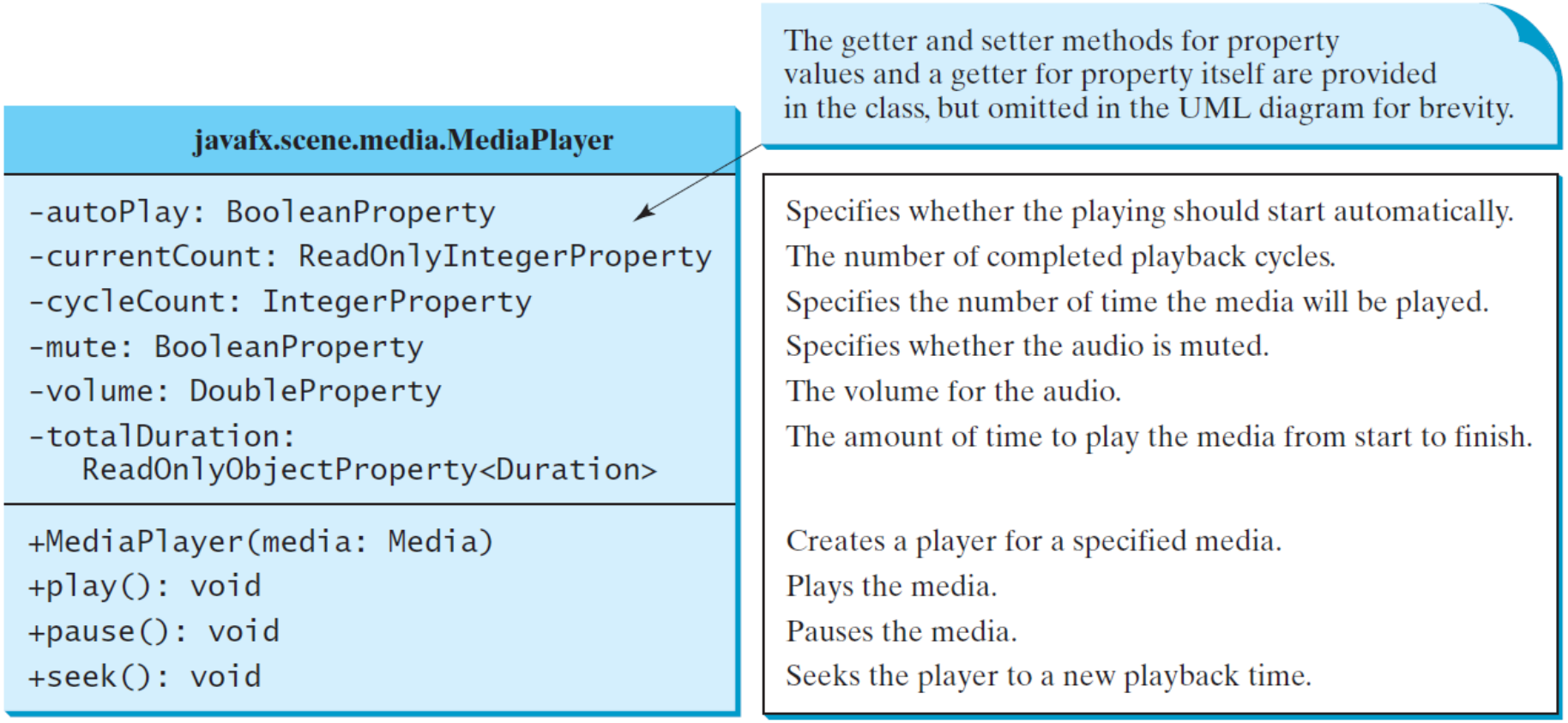
Media

You can use the **Media** class to obtain the source of the media, the **MediaPlayer** class to play and control the media, and the **MediaView** class to display the video.



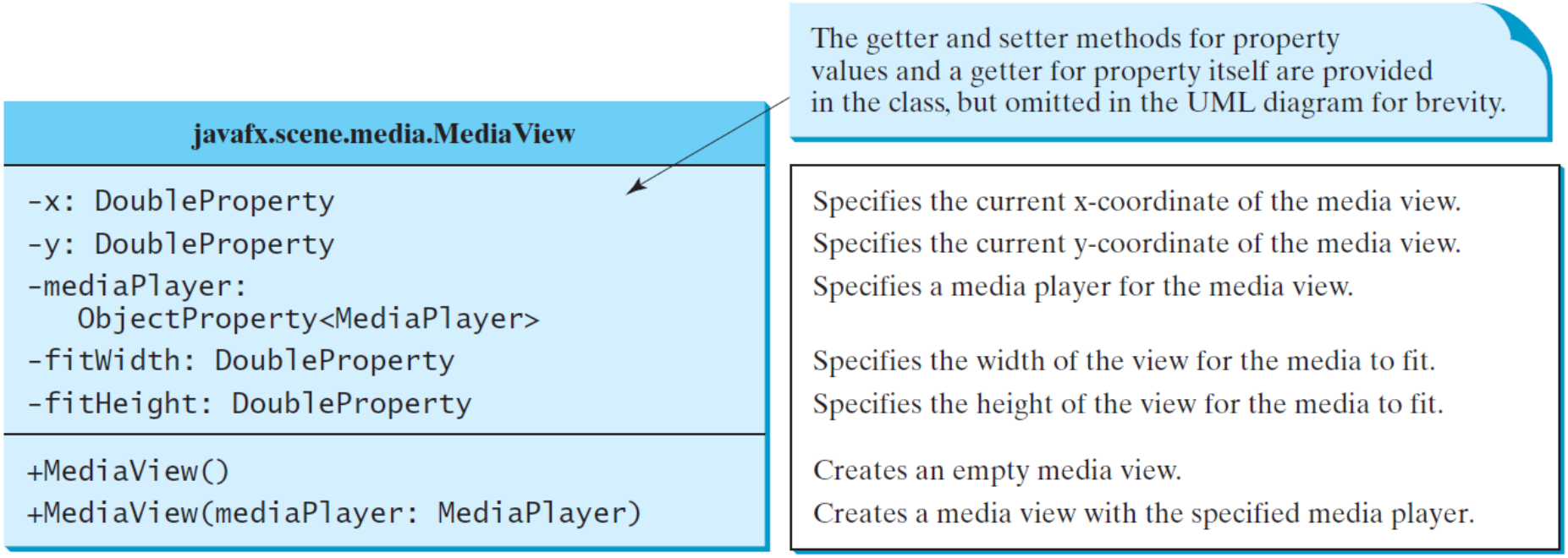
MediaPlayer

The **MediaPlayer** class playes and controls the media with properties such as **autoPlay**, **currentCount**, **cycleCount**, **mute**, **volume**, and **totalDuration**.



MediaView

The **MediaView** class is a subclass of **Node** that provides a view of the **Media** being played by a **MediaPlayer**. The **MediaView** class provides the properties for viewing the media.

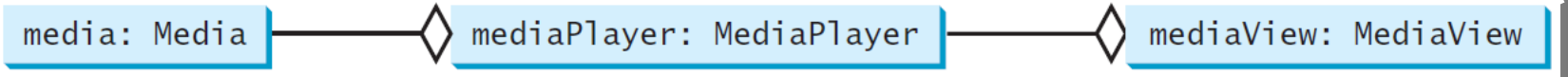


Liang, Introduction to Java Programming, Eleventh Edition, (c) 2018 Pearson Education, Ltd. All rights reserved.

Example: Using Media

This example displays a video in a view. You can use the play/pause button to play or pause the video and use the rewind button to restart the video, and use the slider to control the volume of the audio.

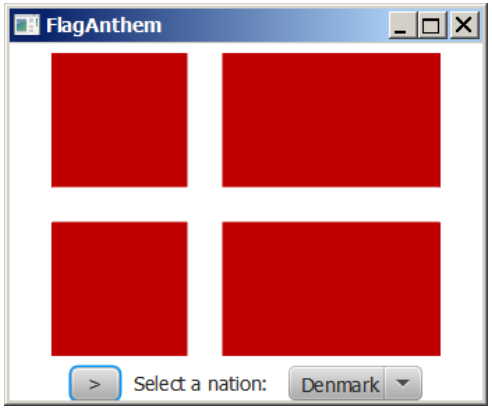




MediaDemo Run

Case Study: National Flags and Anthems

This case study presents a program that displays a nation's flag and plays its anthem.







FlagAnthem Run

Liang, Introduction to Java Programming, Eleventh Edition, (c) 2018 Pearson Education, Ltd. All rights reserved.